Carbon Calculator App: Methodology

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The repository for the carbon calculator, including installation steps, is available at: https://github.com/tejas-n-rao/EE_CreativeProject. A deployed web version is available at: https://ee-creative-project-web.vercel.app/.

1 Scope and Reporting Units

This document explains the implemented methodology for estimating monthly household greenhouse gas emissions and for calculating comparative carbon index scores.

- The primary reported output is monthly emissions in kilograms of carbon dioxide equivalent ($kgCO_2e$/month).
- Annual emissions are computed only to compare the user against annual per-capita benchmarks.
- Benchmark values are expressed in tonnes of carbon dioxide equivalent per person per year (tCO_2e/person/year).

2 Definitions, Abbreviations, and Notation

2.1 Core Terminology

- **Activity line**: one row of user activity data, identified by activity category, unit, region code, and date context.
- **Emission factor**: the amount of greenhouse gas emissions attributed to one unit of activity (for example, $kgCO_2e$ per kilometre).
- **Accessible weekly form**: a simplified questionnaire that collects weekly household behaviour and maps it to estimated monthly activity values.
- **Fallback factor**: a factor used when a region-specific value is unavailable; this implementation falls back to the global (WORLD) value.

2.2 Abbreviations

Abbreviation	Meaning
API	Application Programming Interface
GHG	Greenhouse gas

Abbreviation	Meaning
$kgCO_2e$	Kilograms of carbon dioxide equivalent
tCO_2e	Tonnes of carbon dioxide equivalent
LCA	Life cycle assessment
LPG	Liquefied petroleum gas
NCV	Net calorific value
GWP	Global warming potential
CO_2	Carbon dioxide
CH_4	Methane
N_2O	Nitrous oxide
kWh	Kilowatt-hour
AC	Air conditioner (or air conditioning use, depending on context)
URL	Uniform Resource Locator
IN	Region code representing India
WORLD	Region code representing a global average or global proxy
CEA	Central Electricity Authority (India)
India GHGP	India Greenhouse Gas Program
TJ	Terajoule
kt	Kilotonne
t	Tonne (metric tonne)
km	Kilometre
L	Litre
m^3	Cubic metre

2.3 Notation

Symbol	Meaning	Unit
i	Activity-line index used in aggregation across user inputs	Dimensionless
A_i	Activity quantity for activity line i	Activity-dependent
EF_i	Emission factor for activity line i	$kgCO_2e$ per activity unit
E_i	Emissions from activity line i	$kgCO_2e$/month
E_{monthly}	Total monthly emissions across all activity lines	$kgCO_2e$/month
$E_{\text{annual, kg}}$	Annual emissions in kilograms	$kgCO_2e$/year
$E_{\text{annual, t}}$	Annual emissions in tonnes	tCO_2e/year
B_{India}	India per-capita benchmark value	tCO_2e/person/year
B_{World}	World per-capita benchmark value	tCO_2e/person/year
$E_{\text{class, monthly, kg}}$	Monthly category subtotal used for dashboard class comparisons	$kgCO_2e$/month
$E_{\text{class, annual, t}}$	Annualized category subtotal used for dashboard class comparisons	tCO_2e/year
$I_{\text{refrigerator}}$	Refrigerator indicator: 1 if present, 0 if absent	Dimensionless
j	Food item index in diet-factor summation	Dimensionless

Symbol	Meaning	Unit
$\text{intake}_{j,\text{kg/day}}$	Daily mass intake of food item j	kg/day
$EF_{j,\text{kgCO}_2\text{e/kg}}$	Food-level emission factor for item j	$\text{kgCO}_2\text{e/kg}$
EF_{CO_2}	Carbon dioxide emission factor used in the LPG derivation	t/TJ
EF_{CH_4}	Methane emission factor used in the LPG derivation	t/TJ
$EF_{\text{N}_2\text{O}}$	Nitrous oxide emission factor used in the LPG derivation	t/TJ
GWP_{CH_4}	Global warming potential multiplier for methane	Dimensionless
$GWP_{\text{N}_2\text{O}}$	Global warming potential multiplier for nitrous oxide	Dimensionless

3 Implemented Calculation Pipeline

3.1 Step 1: Input Capture and Normalization

The calculator supports two user input modes:

- **Monthly form**: the user enters monthly activity values directly.
- **Accessible weekly form**: the user enters weekly behavioural values; these are converted to monthly quantities before submission to the Application Programming Interface (API).

General weekly-to-monthly conversion:

$$\text{Monthly value} = \text{Weekly value} \times 4.345$$

The multiplier 4.345 is the average number of weeks per month.

For the accessible weekly form, additional proxy conversions are applied:

Flight activity conversion:

$$\text{Monthly flight passenger-km} = \frac{\text{Flights per year} \times \text{Average flight distance (km)}}{12}$$

Electricity proxy:

$$\text{Electricity}_{\text{kWh/month}} = 45 + 18 \cdot \text{household size} + 1.2 \cdot 30 \cdot \text{AC hours/day} + 25 \cdot I_{\text{refrigerator}}$$

Water proxy (first weekly litres, then monthly cubic metres):

$$\text{Weekly water (L)} = 40 \cdot 7 \cdot \text{household size} + 50 \cdot \text{showers/week} + 70 \cdot \text{laundry loads/week}$$

$$\text{Water}_{\text{m}^3\text{/month}} = \frac{\text{Weekly water (L)} \times 4.345}{1000}$$

Liquefied petroleum gas (LPG) cooking proxy:

$$\text{LPG}_{\text{kg/month}} = \text{home-cooked meals/week} \times 0.08 \times 4.345$$

Diet activity conversion:

$$\text{Diet activity} = 30 \text{ person-day/month}$$

3.2 Step 2: Emission Factor Resolution

For each activity line (category, unit, region, date), the API resolves an emission factor from `data/emission_factors.xlsx`. If a matching region-specific factor is unavailable, the implementation uses the WORLD fallback factor.

3.3 Step 3: Line Emissions and Aggregation

For each activity i:

$$E_i = A_i \times EF_i$$

The monthly total is:

$$E_{\text{monthly}} = \sum_i E_i$$

Implemented rounding protocol:

- Each line emission E_i is rounded to 4 decimal places.
- The total is computed from these rounded line values.
- The final total is then rounded to 4 decimal places.

3.4 Step 4: Annualization and Carbon Index

Annualization is computed as:

$$E_{\text{annual, kg}} = 12 \cdot E_{\text{monthly}}$$

$$E_{\text{annual, t}} = \frac{E_{\text{annual, kg}}}{1000}$$

Index scores are computed relative to per-capita benchmark values:

$$\text{India Index} = \frac{E_{\text{annual, t}}}{B_{\text{India}}} \times 100$$

$$\text{World Index} = \frac{E_{\text{annual, t}}}{B_{\text{World}}} \times 100$$

Both index scores are rounded to 2 decimal places.

3.5 Step 5: Class-Level Benchmark Comparison (Dashboard)

For dashboard benchmarking, activity lines are grouped into class categories:

- **Transportation**: petrol car, diesel car, two-wheeler, bus, metro, rail, short-haul flight.
- **Food**: diet category emissions and LPG cooking emissions.
- **Water**: water supply emissions.

Category-level annualization is computed by:

$$E_{\text{class, annual, t}} = \frac{12 \cdot E_{\text{class, monthly, kg}}}{1000}$$

4 Emission Factor Table

This table lists the implemented emission factors used by the calculator.

Unit notes for coded units used in this table:

- `person_day` means one person following a diet pattern for one day.
- `passenger_km` means one passenger travelling one kilometre.

Activity	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Bus	WORLD	km	0.097	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diesel Car	WORLD	km	0.171	Our World in Data, 2025	https://ourworldindata.org/travel-carbon-footprint
Diet (Meat-heavy)	WORLD	person_day	4.850	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Mixed)	WORLD	person_day	2.750	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Diet (Plant-based)	WORLD	person_day	1.800	Custom, 2025	https://www.bcia.com/sites/default/files/imce/Branch%20PD/Vic%26Islands%20branch/Poore-Nemecek-2018.pdf
Electricity	IN	kWh	0.757	CEA, 2025	https://www.climatiq.io/data/emission-factor/34488b8a-0e4a-47e1-a309-8a15c utm_source=chatgpt.com
Short-haul Flight	IN	passenger_km	0.121	India GHGP, 2014	https://indiaghgp.org
Cooking Gas (LPG)	IN	kg	2.990	Custom, 2025	https://wgbis.ces.iisc.ac.in/energy/paper/GHG_footprint/RSER_bharath2015.pdf

Activity	Region	Unit	Factor ($kgCO_2e$/unit)	Source (Name, Year)	Source URL
Metro	WORLD	km	0.041	Custom, 2025	https://www.researchgate.net/publication/300425890_Energy_use_and_carbon_dioxide_emissions_assessment_in_the_lifecycle_of_passenger_rail_systems_The_case_of_the_Rio_de_Janeiro_Metro
Petrol Car	WORLD	km	0.171	Our World in Data, India GHGP, 2025	https://ourworldindata.org/travel-carbon-footprint
Rail	IN	km	0.035	India GHGP, 2025	https://indiaghgp.org
Two-wheeler	IN	km	0.035	India GHGP, 2025	https://www.researchgate.net/publication/343818309_India_Specific_Road_Transport_Emission_Factors
Water Supply	WORLD	m^3	0.344	UK Environment Agency, 2025	(URL not provided in seed data)

5 Custom Factor Derivation Methods

For selected activities, a direct factor in $kgCO_2e$ per implemented unit was not available in the seed dataset. In those cases, the repository uses documented custom derivations.

5.1 Diet Factors

The repository note `data/Custom_Estimation.md` derives diet factors from food-level life cycle assessment (LCA) intensities, primarily based on Poore and Nemecek (2018).

General diet construction equation:

$$EF_{\text{diet}} = \sum_j \left(\text{intake}_{j,\text{kg/day}} \times EF_{j,\text{kgCO}_2\text{e/kg}}\right)$$

Raw archetype outputs documented in the repository:

$$EF_{\text{meat-heavy, raw}} = 11.9 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{mixed, raw}} = 5.85 \text{ kgCO}_2\text{e/day}$$

$$EF_{\text{plant-based, raw}} = 2.85 \text{ kgCO}_2\text{e/day}$$

After practical adjustment steps (food waste allowance, edible-portion correction, retail-to-consumption mass correction, and caloric normalization), the implemented factors are:

$$EF_{\text{diet, meat-heavy}} = 4.85 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, mixed}} = 2.75 \text{ kgCO}_2\text{e/person-day}$$

$$EF_{\text{diet, plant-based}} = 1.80 \text{ kgCO}_2\text{e/person-day}$$

5.2 LPG Factor

The LPG factor is derived using net calorific value (NCV), pollutant emission factors, and global warming potential (GWP) multipliers.

General equation:

$$EF_{\text{fuel, kgCO}_2\text{e/kg}} = \text{NCV}\,[EF_{\text{CO}_2} + EF_{\text{CH}_4} \cdot GWP_{\text{CH}_4} + EF_{\text{N}_2\text{O}} \cdot GWP_{\text{N}_2\text{O}}]$$

Repository constants:

- NCV = 47.3 TJ/kt fuel
- $EF_{\text{CO}_2} = 63.1$ t/TJ, $EF_{\text{CH}_4} = 0.005$ t/TJ, $EF_{\text{N}_2\text{O}} = 0.0001$ t/TJ
- $GWP_{\text{CH}_4} = 28$, $GWP_{\text{N}_2\text{O}} = 265$

Computed components (as implemented in the repository):

$$\text{CO}_2 \text{ component} = 2.9836 \text{ kgCO}_2\text{e/kg}$$

$$\text{CH}_4 \text{ component (as CO}_2\text{e)} = 0.00662 \text{ kgCO}_2\text{e/kg}$$

$$\text{N}_2\text{O component (as CO}_2\text{e)} = 0.00125 \text{ kgCO}_2\text{e/kg}$$

$$EF_{\text{LPG}} = 2.9836 + 0.00662 + 0.00125 \approx 2.99 \text{ kgCO}_2\text{e/kg}$$

5.3 Metro

The custom note documents an indicative operational range of approximately 0.03 to 0.05 kgCO_2e per passenger-kilometre for electric urban rail systems, and then selects a midpoint-style value for implementation.

$$EF_{\text{metro}} = 0.041 \text{ kgCO}_2\text{e/km}$$

In this implementation, one travelled kilometre in the metro activity input is interpreted as one passenger-kilometre.

6 Benchmark Dataset Used for Indexing

The API uses the following annual per-capita greenhouse gas (GHG) benchmarks:

- India benchmark: 2.9377 tCO_2e/person/year (2024).
- World benchmark: 6.6691 tCO_2e/person/year (2024).

Additional category-level class benchmark statistics (transportation, food, and water) are used in the dashboard and are seeded in `data/db_benchmark_stats.json`.

7 Limitations

- The model uses direct activity-by-factor multiplication and does not include many second-order life cycle effects.
- Several factors are scenario-dependent custom estimates, especially for diet and LPG.
- Region-specific coverage is incomplete; when required, WORLD fallback values are applied.
- Benchmark datasets are drawn from different years across metrics, which limits strict cross-metric comparability.
- Accessible weekly form utility proxies are heuristic and may differ from metered household records.

Technology Stack

- **Backend**: FastAPI, SQLAlchemy object-relational mapping (ORM), Alembic migrations, Pydantic, PostgreSQL/SQLite databases.
- **Frontend**: Next.js (Application Router), React, TypeScript, and custom Cascading Style Sheets (CSS).
- **Data assets**: JavaScript Object Notation (JSON) seed files and an editable spreadsheet factor sheet (`data/emission_factors.xlsx`).
- **Tooling**: Pytest, Node Package Manager (npm) workspaces, Docker Compose, ESLint, and Prettier.